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Tretinoin 0.1% and Benzoyl Peroxide 3% Cream for the Treatment of Facial Acne Vulgaris

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Abstract

Objective: To assess the efficacy, safety, and clinical application of tretinoin 0.1%-benzoyl peroxide 3% cream for the topical treatment of acne vulgaris. **Data Sources:** A systematic review of the literature was performed using the terms *Twyneo* OR *tretinoin and benzoyl peroxide* OR *S6G5T-3* in MEDLINE (PubMed) and EMBASE. ClinicalTrials.gov was searched to obtain completed clinical trial results not published elsewhere. **Study Selection and Data Extraction:** All human studies published in English prior to November 2022 related to pharmacology, clinical trials, safety, and efficacy were evaluated for inclusion. **Data Synthesis:** In two 12-week, phase 3, randomized, vehicle-controlled clinical trials, tretinoin 0.1%-benzoyl peroxide 3% cream significantly reduced inflammatory and noninflammatory facial acne lesions and significantly improved Investigator Global Assessment (IGA) rating to clear or almost clear. The cream has a suitable safety profile, with application site pain and dryness as the most common adverse events. **Relevance to Patient Care and Clinical Practice in Comparison to Existing Agents:** Tretinoin-BPO had similar IGA success compared to other topical retinoid and retinoid-BPO treatments for acne vulgaris. Compared to individual tretinoin and benzoyl peroxide therapy, the combination product streamlines application, which will improve medication adherence; however, the cost of tretinoin-BPO cream may be prohibitive. **Conclusions:** Tretinoin 0.1%-benzoyl peroxide 3% cream is safe and effective for the treatment of moderate-to-severe acne. Long-term trial data on efficacy and tolerability are not yet available.

Keywords
acne, adherence, microencapsulation, patient preferences, topical therapy

Introduction

Acne vulgaris is a relapsing-remitting inflammatory disorder of the pilosebaceous unit caused by androgen-induced excess sebum production, aberrant desquamation of the sebaceous follicle, and *Propionibacterium acnes* proliferation.[1,2] Approximately 20% of young people are affected by moderate-to-severe acne. Complications include scarring, infection, and reduced quality of life impacted by depression, anxiety, and social withdrawal.[3] Topical acne treatments include retinoids, benzoyl peroxide, clindamycin, and beta hydroxy acids, used alone or in combination.[3] According to the most recent clinical guidelines, combination use of topical retinoids and benzoyl peroxide are first line for moderate to severe acne.[4,5] Topical retinoids help eliminate comedones, while benzoyl peroxide offers broad-spectrum anti-infective effects.

Adherence to topical acne treatment is a major hurdle to successful treatment outcomes.[6] While combinations of different topicals are often prescribed to address the multifactorial nature of acne, the complexity of these regimens further impairs treatment adherence.[7] Combination drug products allow for simpler medication regimes, which can help improve adherence compared to the use of multiple separate agents.[7] While some retinoids, such as adapalene and tazarotene, exist in stable form with benzoyl peroxide (BPO), BPO can oxidize tretinoin, which

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Table 1. Plasma Concentrations of Tretinoin and its Metabolites When Tretinoin 0.1%-Benzoyl Peroxide 3% Cream Was Used for the Treatment of Acne Vulgaris.

Compound	Age	Mean (±SD) C_{max} ($\frac{ng}{mL}$)	Mean (±SD) $AUC_{0\text{-}24}$ ($\frac{ng \cdot h}{mL}$)
Tretinoin	9 to 11	0.18 ± 0.22	2.06 ± 3.96
	12 to 17	0.19 ± 0.18	1.56 ± 1.97
	≥18	0.15 ± 0.17	0.63 ± 0.95
4-keto 13-cis RA	9 to 11	0.34 ± 0.36	2.89 ± 3.17
	12 to 17	0.32 ± 0.28	2.39 ± 3.05
	≥18	0.27 ± 0.29	1.99 ± 2.90
13-cis RA	9 to 11	0.15 ± 0.17	0.96 ± 1.26
	12 to 17	0.28 ± 0.35	1.79 ± 2.79
	≥18	0.21 ± 0.19	1.99 ± 2.90

Source: Adapted from Twyneo.[13]
Abbreviations: $AUC_{0\text{-}24}$, area under the plasma concentration-time curve; C_{max}, maximum concentration; SD, standard deviation.

has prevented the development of topical products that consolidate these two standard acne treatments into one.[8]

Microencapsulation provides a potential means to combine drugs that would normally be incompatible. In July 2021, the U.S. Food and Drug Administration (FDA) approved tretinoin 0.1%-benzoyl peroxide 3% cream as a once-daily application for moderate-to-severe acne vulgaris. While a tretinoin microsphere produce exists, this is the first FDA-approved tretinoin-benzoyl peroxide combination therapy for the treatment of acne.[9]

The purpose of this review is to assess the efficacy, safety, and clinical application of tretinoin 0.1%-benzoyl peroxide 3% cream in the treatment of acne vulgaris.

Methods

A systematic review of the literature was performed using the terms *Twyneo* OR *S6G5T3* OR *tretinoin AND benzoyl peroxide* in MEDLINE (PubMed) and EMBASE. ClinicalTrials.gov was searched to obtain completed clinical trial results not published elsewhere. All human studies published in English before November 2022 related to pharmacology, phase III clinical trials, safety, and efficacy were evaluated for inclusion.

Results

Drug Pharmacology

Mechanism of action. Benzoyl peroxide and tretinoin have distinct but complementary mechanisms of action. BPO is an oxidizing agent. It penetrates the pilosebaceous unit and generates free radicals that damage the cell wall of *Propionibacterium acnes* upon application to the skin.[10] BPO also has mild comedolytic and anti-inflammatory properties.[10]

Tretinoin modifies gene transcription and expression by binding to retinoic acid receptors (RAR) alpha, beta, gamma, and retinoid X receptors in keratinocytes. Although its mechanism is not entirely understood, tretinoin likely promotes epidermal cell proliferation, cornified cell detachment, and shedding. The increased follicular epithelial cell turnover and elimination of sebum in sebaceous ducts facilitate the extrusion of comedones.[10]

Encapsulation. Tretinoin and BPO are effective in the treatment of inflammatory acne.[11] Guidelines recommend individual use of each medication because benzoyl peroxide can oxidize tretinoin.[5] However, tretinoin 0.1%-benzoyl peroxide 3% cream was developed using microencapsulation. Microencapsulation separates BPO and tretinoin in amorphous silica shells to protect tretinoin from BPO-induced oxidation.[12] This allows for the delivery of both stable ingredients in their product.[8] Microencapsulation technology also allows for delayed medication release onto the skin, reducing the potential for irritation seen with tretinoin and BPO and increasing patient tolerability.[8,10]

Pharmacokinetics data. An open-label study of 35 participants nine years of age and older with acne vulgaris was conducted to evaluate systemic absorption. Participants applied 1.9 grams of the tretinoin 0.1%-benzoyl peroxide 3% cream once daily to the skin of the face, upper chest, shoulders, and upper back for two weeks. Serum was drawn on day 14 to determine the mean peak serum concentration (C_{max}) and area under the plasma concentration-time curve ($AUC_{0\text{-}24}$) of tretinoin and its metabolites (Table 1).[9] Across all age groups, the mean peak serum concentration of tretinoin and its metabolites did not reach levels of systemic toxicity in patients aged nine and older. Topical benzoyl peroxide is absorbed by the skin and converted to benzoic acid, which is rapidly eliminated in the urine, providing no risk for systemic absorption.[13,14]

Table 2. Demographic and Baseline Acne Vulgaris Characteristics.

	Clinical trial[15]		Clinical trial[16]	
	Tretinoin-BPO (n = 281)	Vehicle (n = 143)	Tretinoin-BPO (n = 290)	Vehicle (n = 144)
Demographics	—	—	—	—
Age (years), mean	20.9	21.4	20.1	20.3
Gender (F/M), % (n)	62.3%/37.7% (175/106)	58.0%/42.0% (83/60)	59.7%/40.3% (173/117)	53.5%/46.5% (77/67)
Ethnicity, % (n)	—	—	—	—
Hispanic or Latino	36.3% (102)	30.7% (44)	29.3% (85)	38.9% (56)
Not Hispanic or Latino	63.3% (178)	68.5% (98)	70.3% (204)	60.4% (87)
Unknown	0.4% (1)	0.7% (1)	0.3% (1)	0.7% (1)
Race, % (n)	—	—	—	—
American Indian/Alaska native	0.4% (1)	0.7% (1)	0.7% (2)	1.4% (2)
Asian	9.3% (26)	7.0% (10)	4.1% (12)	4.2% (6)
Native Hawaiian/Pacific Islander	0.4% (1)	1.4% (2)	0.3% (1)	0.7% (1)
Black	18.9% (53)	14.0% (20)	17.9% (52)	12.5% (18)
White	69.0% (194)	76.2% (109)	73.1% (212)	76.4% (110)
More than one race	2.1% (6)	0.7% (1)	3.8% (113)	4.9% (7)
Unknown or not reported	0.0% (0)	0.0% (0)	0.0% (0)	0.0% (0)
Baseline Facial Acne Vulgaris	—	—	—	—
Characteristics	—	—	—	—
IGA, % (n)	—	—	—	—
3—Moderate	89.3% (251)	92.3% (132)	90.3% (262)	92.4% (133)
4—Severe	10.7% (30)	7.7% (11)	9.7% (28)	6.9% (10)
Inflammatory acne lesion count, #	33.5	33.5	28.2	27.5
Noninflammatory acne lesion count, #	48.6	47.1	44.6	44.9

Abbreviations: BPO, benzoyl peroxide; IGA, investigator global assessment.

Summary of Clinical Trials for Acne Vulgaris Treatment

The safety and efficacy of tretinoin 0.1%-benzoyl peroxide 3% cream were evaluated in two 12-week, phase 3, double-blind, randomized, vehicle-controlled trials, although these data are yet to be published in a peer-reviewed journal.[15,16] Patients 9 years of age and older with moderate to severe facial acne vulgaris, defined as an Investigator Global Assessment (IGA) greater than or equal to 3 (where 0 = *clear*, 1 = *almost clear*, 2 = *mild acne*, 3 = *moderate acne*, 4 = *severe acne*), were included. Patients with more than two acne nodules or cysts, acne conglobate, acne fulminans, secondary acne, history of blood dyscrasias, or those with an underlying disease requiring topical or systemic therapy that might interfere with study assessments were excluded.[15,16]

Patient characteristics were balanced across age, gender, race, ethnicity, baseline IGA, and baseline facial noninflammatory and inflammatory acne lesions (Table 2). Primary outcomes included IGA success and a mean reduction in facial inflammatory and noninflammatory lesions at week 12 compared to baseline. IGA success was defined as an IGA score of clear (score=0), defined as normal skin with no evidence of acne vulgaris, or almost clear (score=1), defined as rare noninflammatory lesions with rare noninflamed papules, and at least a two-grade improvement from the patient's baseline score.[17] Inflammatory lesions were defined as papules, pustules, nodules, and cysts.[15,16] Noninflammatory lesions were defined as open and closed comedones.[15,16] IGA success was reported as a percentage of participants achieving IGA success. The change from baseline inflammatory and noninflammatory lesions was calculated using least squares means and standard deviations from an analysis of covariance (ANCOVA). Covariates included treatment group, analysis center, and baseline lesion count.[15,16] For all analyses, an intent-to-treat (ITT) population was employed, including all participants who were randomized and dispensed a study drug.[15,16]

In the first clinical trial, 424 subjects were randomized into one of two arms, treatment with (1) tretinoin-BPO cream or (2) vehicle cream; 249 (88.6%) of 281 and 131 (91.6%) of 143 treated with tretinoin-BPO cream and vehicle cream, respectively, completed the study.[15] In the second clinical trial, 434 subjects were randomized into (1) tretinoin-BPO cream or (2) vehicle cream; 242 (83.4%) of 290 and 131 (91.6%) of 144 treated with tretinoin-BPO cream and vehicle cream, respectively, completed the study.[16]

The 12-week IGA success rates in the two studies were 38.5 and 11.5% ($P < 0.001$)[15,17] and 25.4 and 14.7% ($P = 0.017$)[16,17] for tretinoin-BPO cream and vehicle cream, respectively.[15-17] Mean reductions in inflammatory lesions from baseline were –21.6 and –14.8 ($P < 0.001$)[15,17] and –16.2 and –14.1 ($P = 0.018$)[16,17] for tretinoin-BPO cream

Table 3. Adverse Events With Once Daily Tretinoin 0.1%/Benzoyl Peroxide 3% Use for 12 Weeks.

	Clinical trial[15]		Clinical trial[16]	
	Tretinoin-BPO (n = 274)	Vehicle (n = 139)	Tretinoin-BPO (n = 281)	Vehicle (n = 138)
Serious Adverse Events, % (n)	—	—	—	—
Total	0.00% (0)	0.00% (0)	0.36% (1)	0.72% (1)
Depression	—	—	0.36% (1)	0.72% (1)
Bipolar II Disorder	—	—	0.00% (0)	0.72% (1)
Conduct Disorder			0.00% (0)	0.72% (1)
Adverse Events, % (n)	—	—	—	—
General Disorders	—	—	—	—
Application Site Pain	12.04% (33)	0.00% (0)	9.61% (27)	0.72% (1)
Application Site Dryness	6.20% (17)	0.72% (1)	3.56% (10)	0.00% (0)
Application Site Erythema	5.11% (14)	0.00% (0)	2.85% (8)	0.00% (0)
Application Site Exfoliation	5.11% (14)	0.00% (0)	3.20% (9)	0.00% (0)
Application Site Dermatitis	1.46% (4)	0.72% (1)	1.78% (5)	0.00% (0)
Application Site Irritation	1.46% (4)	0.00% (0)	—	—
Application Site	—	—	1.07% (3)	0.72% (1)
Photosensitivity	—	—	—	—
Infections	2.19% (6)	2.16% (3)	0.36% (1)	1.45% (2)
URI	—	—	1.42% (4)	0.00% (0)
Nasopharyngitis	—	—	—	—
Nervous System Disorders	1.09% (3)	0.72% (1)	—	—
Headache	—	—	—	—
Psychiatric Disorders	—	—	—	—
ADHD	0.00% (0)	1.44% (2)	—	—

Abbreviations: ADHD, attention deficit/hyperactivity disorder; BPO, benzoyl peroxide; URI, upper respiratory tract infection.

and vehicle cream, respectively.[15-17] Mean reductions in noninflammatory lesions from baseline were –29.7 and –19.8 ($P < 0.001$)[15,17] and –24.2 and –17.4 ($P < 0.001$)[16,17] for tretinoin-BPO cream and vehicle cream, respectively.[15-17]

In the first clinical trial,[15] 25.2% versus 5.8% and in the second clinical trial,[16] 24.2% versus 2.9% treated with tretinoin-BPO cream versus vehicle cream reported adverse events, respectively. Participants reported application site pain (9.6%-12.0%), dryness (3.6%-6.2%), erythema (2.9%-5.1%), dermatitis (1.5%-1.8%), exfoliation (3.2%-5.1%), irritation (1.5%),[15] and photosensitivity (1.1%).[16] One participant (0.36%)[15] reported depression (Table 3).[15,16]

Relevance to Patient Care and Clinical Practice in Comparison to Existing Drugs

Benzoyl peroxide with topical retinoids is indicated for the treatment of moderate to severe acne vulgaris.[4,5] Tretinoin 0.1%-benzoyl peroxide 3% cream is the first and only BPO and tretinoin combination cream approved for once-daily application in patients with acne. Its microencapsulation technology permits the simultaneous use of both active ingredients, which was previously not possible given BPO-induced degradation of tretinoin. Combining BPO's bactericidal effects and tretinoin's increased follicular epithelial cell turnover, tretinoin-BPO targets both *Propionibacterium acnes* colonization acnes and follicular hyperkeratinization.

The clinical significance of the simultaneous application of BPO and retinoids using microencapsulation is uncertain. Tretinoin-BPO had similar IGA success compared to other retinoids and retinoid-BPO combinations for acne vulgaris (Table 4).[18,19,21] However, head-to-head comparisons between tretinoin-benzoyl peroxide cream and other medications approved for moderate-to-severe acne are not available.

A limitation to patient medication use is cost. One 30-g cream pump of tretinoin-benzoyl peroxide cream costs approximately $457.[20] A manufacturer discount may reduce the cost to $0 for patients with insurance and $60 for patients without insurance, which may improve access.[22] A limitation to all topical medication regimens is adherence. Patients are poorly adherent to topical medications due to inconvenience, discomfort, and complex treatment regimens.[23] Tretinoin-benzoyl peroxide cream removes the barriers patients face to multi-step topical acne treatment by combining medication steps into one easy-to-use cream.

Table 4. IGA Success of Acne Vulgaris Trials.

Drug	Tretinoin 0.1%/BPO 3% cream[15,17]	Tretinoin 0.1%/BPO 3% cream[16,17]	Adapalene 0.1% gel[18]	Adapalene 0.3% gel[18]	Adapalene 0.3%/BPO 2.5% gel[19]	Trifarotene 000.5% cream[20]
IGA Success	38.5%**	25.4%*	16.9%**	23.3%**	31.9%*	29.4%-42.3%**
Vehicle Success	11.5%	14.7%	10%	10%	11.8%	19.5%-25.7%
Absolute IGA success	27%	10.7%	7%**	12%	20.1%	9.9%-16.6%

Abbreviations: BPO, benzoyl peroxide; IGA, investigator global assessment.
Significance between IGA success and vehicle success, where *$P < 0.05$ and **$P < 0.01$.

Current clinical guidelines recommend tretinoin and BPO be applied at different times, given concerns that tretinoin may be inactivated by the coadministration of benzoyl peroxide.[5] It is typically recommended that BPO be used in the morning and tretinoin in the evening, as some tretinoin formations are not photostable.[5] Tretinoin 0.1%-benzoyl peroxide 3% cream circumvents these obstacles as it is a once daily application that is photostable.[24] The combination product streamlines application, which will improve medication adherence; however, the cost of tretinoin-BPO cream may be prohibitive. Without insurance or the manufacturers discount, the combination medication may cost up to $460. Conversely, a 20-g tube of tretinoin 0.1% cost approximately $35, while over the counter 2.5%, 5%, and 10% formulations of BPO cost approximately $10.[25,26] Individual therapy is best suited for patients who can adhere to twice daily topical applications. Patients who struggle with twice daily application may have better adherence and clinical response to combination tretinoin-BPO cream.

Tretinoin-benzoyl peroxide cream has a relatively safety profile. Application site pain and dryness were the most common reported AEs. These adverse effects were observed in clinical trials and may not reflect real world clinical practice as patients are less adherent with topical therapy outside the structured environment of a clinical trial.[27]

Conclusion

Tretinoin 0.1% and benzoyl peroxide 3% cream is approved for the treatment of moderate-to-severe acne vulgaris in patients 9 years of age or older. Acne clinical guidelines recommend combining topical retinoids and benzoyl peroxide for moderate to severe acne.[4,5] Tretinoin-benzoyl peroxide cream's microencapsulation technology allows for the combined once-daily application of these ingredients. This simpler medication regime is likely to increase patient adherence to topical therapy and improve treatment outcomes.[7] Tretinoin-benzoyl peroxide cream also demonstrated favorable efficacy safety profiles. Long-term and comparative trials are needed to better characterize the product.

Declaration of Conflicting Interests

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